Probability I (SI 427)

Midsem, Full marks: 30
Date: September 21, 2024
Time: 1:30 p.m. – 3:30 p.m.

1. Prove or disprove the following statement: [12]

   (a) If $\mathcal{F}$ is a field and $A, B \in \mathcal{F}$, then the symmetric difference $A \triangle B \in \mathcal{F}$. ✓

   (b) $\mathcal{G} = \{A \subset \mathbb{N} : A \text{ is a finite set}\}$ is a field. No

   (c) If $\mathcal{F}_1$ and $\mathcal{F}_2$ are sigma-fields, then $\mathcal{F}_1 \cup \mathcal{F}_2$ is a sigma-field. No

   (d) If P_1 and P_2 are probability measures on $(\Omega, \mathcal{F})$, then P, where $P(A) = \alpha P_1(A) + \beta P_2(A)$, is also a probability measure provided $\alpha, \beta \geq 0$ and $\alpha + \beta = 1$. ✓

   (e) Given n events $A_1, A_2, \ldots, A_n$,

   $$\sum_{i=1}^{n} \mathrm{P}(A_i) - \sum_{i<j} \mathrm{P}(A_i \cap A_j) \leq \mathrm{P}\left(\cup_{i=1}^{n} A_i\right).$$ ✓

   (f) If A, B, C are independent events, then A and $C \triangle B$ are independent.

2. Suppose $\{E_n\}_{n\geq 1}$ is a sequence of sets such that $E_1 \subset E_3 \subset E_5 \subset E_7 \cdots$ and $E_2 = E_4 = E_6 = \cdots = E$. Find $\liminf_{n\to\infty} E_n$ and $\limsup_{n\to\infty} E_n$. :E [4]

3. Find distribution function of X, where $X(\omega) = 0$ for $\omega \in A$, $X(\omega) = 1$ for $\omega \in B$, $X(\omega) = 2$ otherwise, and $\mathrm{P}(A) = 1/4$, $\mathrm{P}(B) = 3/4$ and $A \cap B = \emptyset$. [4]
   0, $x<0$; $\frac{1}{4}$, $0\le x<1$; 1, $x\ge 1$

4. A man possesses five coins, one of which is double-headed, one is double-tailed, and three are normal fair coins. He shuts his eyes, picks a coin at random, and tosses it. What is the probability that head appears? [4] $\frac{11}{40}$

5. Let X be independent geometric random variables with parameter p:

   $$\mathrm{P}(X = n) = p(1-p)^{n-1}, \text{ for } n = 1, 2, \ldots.$$

   Define $Y = \min\{X, 100\}$. Find the probability mass function of Y. [3]

6. Find $\mathrm{Var}(X - 2Y)$ where $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$ and $\mathrm{Cov}(X, Y) = 1/2$. 3 [3]

$P(C\cap B^c) = P((B^c\cap C)\cup(C^c\cap B)) = P(B^c\cap C) + P(C^c\cap B)$

$\boxed{A \cap C\triangle B}$

$P(B\cap C^c) = P(B)\,P(C^c)$

$\mathrm{Var}\,X = (E(X^2) - (E(X))^2)$

$\mathrm{Var}\,X = E((X-\mu)^2)$

$\mathrm{Var}(X-2Y) = E((X-2Y)^2) - (E(X-2Y))^2$

$E(X^2) - (E(X))^2$

$\mathrm{Cov}(X,Y) = E((X-EX)(Y-EY))$

$= E(X^2+4Y^2-4XY) - (E(X)-E(2Y))^2$

$= E(XY - XE(Y) - YE(X) + E(X)E(Y))$

$= E(X^2) + E(4Y^2) - E(4XY) - (E(X))^2 - (E(2Y))^2 + 2E(X)E(2Y)$

$= \cancel{E(XY)}$

$= (E(X^2) - E(X)^2) + 4(E(Y^2) - E(Y)^2)$

$1 + 4$

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Quiz 2, Full marks: 10

Date: October 22, 2024

Time: 9:30 a.m. – 10:30 a.m.

1. A group of n people throw their hats into the center of a room. The hats are mixed up, and each person randomly selects one. Let N be the number of people that select their own hats. Fine $E(N)$ and $\operatorname{Var}(N)$. [4]

2. Let X be a random variable and $h : \mathbb{R} \to [0, M]$ be a non-negative function. Show that

$$\frac{E(h(X))}{a} \geq \mathrm{P}(h(X) \geq a) \geq \frac{E(h(X)) - a}{M - a}, \quad \text{whenever} \quad 0 < a < M.$$

[3]

3. Define Cantor set. Show that Cantor set is a Borel set. Find Lebesgue measure of the Cantor set. [3]

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Endsem, Full marks: 50

Date: November 22, 2024. Duration: 3 hours

- Use of any unfair means would lead you to FR grade.
- You have to state the theorems clearly if you are using it in your answer.
- You can not refer/recall solved problems done in class/tutorial session to answer any of the questions.

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1. Suppose $E(X) = E(Y) = 10$, $\text{Var}(X) = 10$, $\text{Var}(Y) = 12$ and $\text{Cov}(X, Y) = -3$. Show that $P(X > Y + 6) < 1/2$. [3]

2. Define Borel σ-algebra on $\mathbb{R}$. Show that $\mathbb{Q}$ (set of rational number) is a Borel set. Find $m([0,1] \setminus \mathbb{Q})$, where m is the Lebesgue measure. [4]

3. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} k(2 - x - y) & \text{for } x > 0, y > 0, x + y < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find k and $E[Y|X = 1]$. [2 + 2]

4. Suppose $X_1 \sim \text{Gamma}(\alpha_1, 1)$, $X_2 \sim \text{Gamma}(\alpha_2, 1)$ and they are independent.

   (a) Find joint density of $(X_1 + X_2, \frac{X_1}{X_1+X_2})$. [2]

   Recall that $X \sim \text{Gamma}(\alpha, \lambda)$ if $f_X(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$

   (b) Is it true that $E\left(\frac{X_1}{X_1 + X_2}\right) = \frac{E(X_1)}{E(X_1 + X_2)}$? Justify your answer. [2]

5. (a) State first Borel Cantelli lemma. [1]

   (b) Let $\{X_n\}$ be a sequence of independent random variables such that $E(X_n) = 0$ and $P(|X_n| < M) = 1$ for all n, where M is a positive real number. Show that $\frac{S_n}{n} \xrightarrow{a.s.} 0$. [4]

6. (a) State Central limit theorem. [1]

   (b) Let $\{X_n\}$ be a sequence of i.i.d. $U(-1, 1)$ random variables. Define

$$Z_n = \sqrt{n}\frac{X_1^3 + X_2^3 + \cdots + X_n^3}{X_1^2 + X_2^2 + \cdots + X_n^2}. \quad N(0, \tfrac{9}{7})$$

   Does Z_n converge in distribution? Justify your answer. If it converges, find the limit. [4]

7. (a) Find MGF of Poisson random variable with parameter λ. [1]